# Timed Automata

# Introduction to TA (1/2)

- Timed Automata support the representation of **concurrent timed** systems
  - Expressivity comparable with that of Time Petri Nets (TPNs)
- A Timed Automaton (TA) is a finite automaton enriched with:
  - a set of **clocks**
  - **timing constraints** on transitions (also termed guards or enabling conditions)
  - **clock resets** on transitions
- A minimal TA: two locations `H` and `K`, two clocks `x` and `y`, a transition from `H` to `K` (enabled if $x > 0$, labeled with action a, resetting clock `y` to 0)

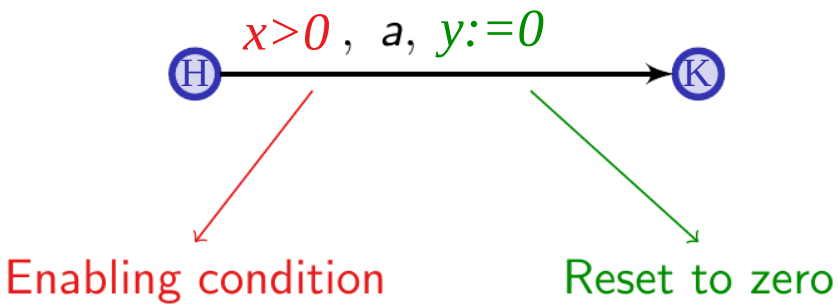
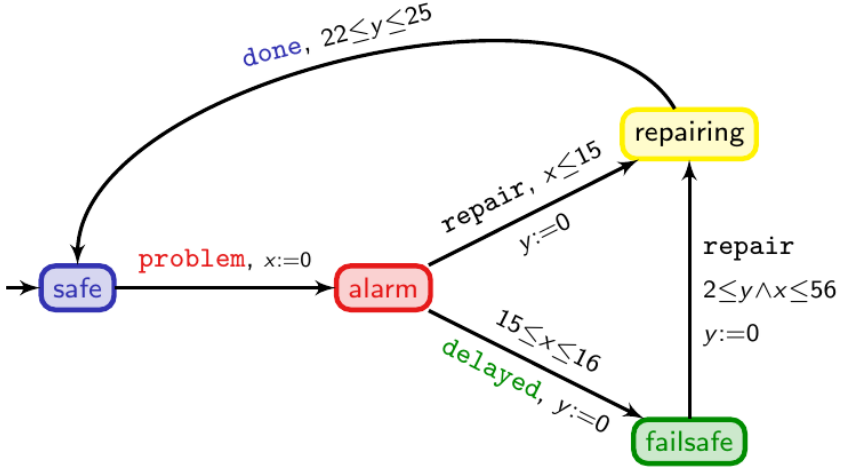



Image adapted from Patricia Bouyer-Decitre, CNRS & ENS Paris-Saclay

R. Alur and D. L. Dill, "A theory of timed automata," Theoretical Computer Science, 1994.

# Introduction to TA (2/2)

- The TA of a repair process




- A possible execution of the TA

| | safe | $\xrightarrow{23}$ | safe | $\xrightarrow{\text{problem}}$ | alarm | $\xrightarrow{15.6}$ | alarm | $\xrightarrow{\text{delayed}}$ | failsafe | |
|---|---|---|---|---|---|---|---|---|---|---|
| x | 0 | | 23 | | 0 | | 15.6 | | 15.6 | ⋯ |
| y | 0 | | 23 | | 23 | | 38.6 | | 0 | |

| | failsafe | $\xrightarrow{2.3}$ | failsafe | $\xrightarrow{\text{repair}}$ | repairing | $\xrightarrow{22.1}$ | repairing | $\xrightarrow{\text{done}}$ | safe |
|---|---|---|---|---|---|---|---|---|---|
| ⋯ | 15.6 | | 17.9 | | 17.9 | | 40 | | 40 |
| | 0 | | 2.3 | | 0 | | 22.1 | | 22.1 |

Image by Patricia Bouyer-Decitre, CNRS & ENS Paris-Saclay

# TA syntax

- A TA is a tuple $\langle L, l_0, \Sigma, X, E \rangle$ where:
  - $L$ is a finite set of **locations**
  - $L_0 \subseteq L$ is the set of **initial location**
  - $\Sigma$ is a finite **action alphabet**
  - $X$ is a set of **clocks**
  - $E \subseteq L \times \Sigma \times G(X) \times 2^X \times L$ is a set of **edges** where $G(X)$ is a set of **guards** with syntax $g := x \sim c \mid g \wedge g$ where $x \in X$, $\sim \in \{<, \leq, =, >, \geq\}$, $c \in \mathbb{N}$ (i.e., a guard is a conjunction of atomic constraints of the form $x \sim c$)
- An example with three locations, two actions, two clocks, and two transitions

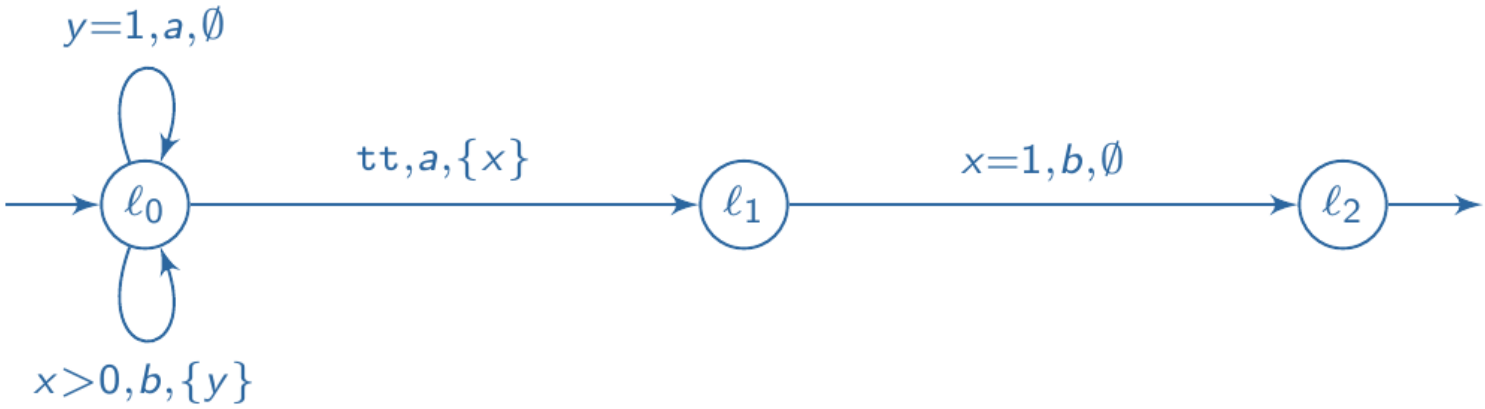



Image by Natalie Bertrand, Inria Rennes Bretagne

# TA semantics (1/3)

- A **clock valuation** $v \in \mathbb{R}_{\geq 0}^{X}$ assigns a value to each clock
  - If a valuation $v \in \mathbb{R}_{\geq 0}^{X}$ satisfies a guard $g \in G(X)$, then we write $v \models g$
  - $v + \tau$ denotes the valuation $(v + \tau)(x) = v(x) + \tau \; \forall x \in X, \forall \tau \in \mathbb{R}_{\geq 0}$
- The **state** of a TA is a pair $(l, v)$ where:
  - $l$ is a location
  - $v$ is a valuation
- There exist two types of **transition** between states
  - **Delay transition**: $(l, v) \xrightarrow{\tau} (l, v + \tau)$ for any state $(l, v)$ and any delay $\tau \in \mathbb{R}_{\geq 0}$
  - **Discrete transition**: $(l, v) \xrightarrow{a} (l', v')$ if $\exists (l, a, g, R, l') \in E$ such that $v \models g$ and $v'(x) = 0$ if $x \in R$ and $v'(x) = v(x)$ if $x \notin R$
- A **run** of a TA is a sequence of alternating delay and discrete transitions: $(l_0, v_0) \xrightarrow{\tau_1} (l_0, v_0 + \tau_1) \xrightarrow{a_1} (l_1, v_1) \xrightarrow{\tau_2} (l_1, v_1 + \tau_2) \xrightarrow{a_2} \ldots \xrightarrow{a_k} (l_k, v_k)$
  - Equivalently written as $(l_0, v_0) \xrightarrow{\tau_1, a_1} (l_1, v_1) \xrightarrow{\tau_2, a_2} \ldots \xrightarrow{\tau_k, a_k} (l_k, v_k)$

# TA semantics (2/3)

- A **time sequence** is a finite non-decreasing sequence of non-negative real values i.e., $s = (t_1)(t_2)\cdots(t_k)$ with $t_i \in \mathbb{R}_{\geq 0}\ \forall\, i \in \{1, \ldots, k\}$
- A **timed word** is a finite sequence of pairs consisting of an action and a time value belonging to a time sequence, i.e., $w = (a_1, t_1)(a_2, t_2)\cdots(a_k, t_k)$ with $a_i \in \Sigma$ and with $(t_1)(t_2)\cdots(t_k)$ being a time sequence
- A timed word $w = (a_1, t_1)(a_2, t_2)\cdots(a_k, t_k)$ is **accepted** by a TA if $\exists\ \rho = (l_0, v_0) \xrightarrow{\tau_1, a_1} (l_1, v_1) \xrightarrow{\tau_2, a_2} \ldots \xrightarrow{\tau_k, a_k} (l_k, v_k)$ with $l_0 \in L_0$ and $t_i = \sum_{1 \leq j \leq i} \tau_j$
- The **accepted timed language** of a TA is the set of its accepted timed words

# TA semantics (3/3)

- A run of the example TA is $(l_0, (0,0)) \xrightarrow{0.1,b} (l_0, (0.1,0)) \xrightarrow{0.2,b} (l_0, (0.3,0)) \xrightarrow{1,a} (l_0, (1.3,1)) \xrightarrow{0.2,b} (l_0, (1.5,0)) \xrightarrow{0,a} (l_1, (0,0)) \xrightarrow{1,b} (l_2, (1,1))$
- An accepted timed word is $w = (b, 0.1)(b, 0.3)(a, 1.3)(b, 1.5)(b, 2.5)$
- Always true guards and empty reset sets are omitted in the representation

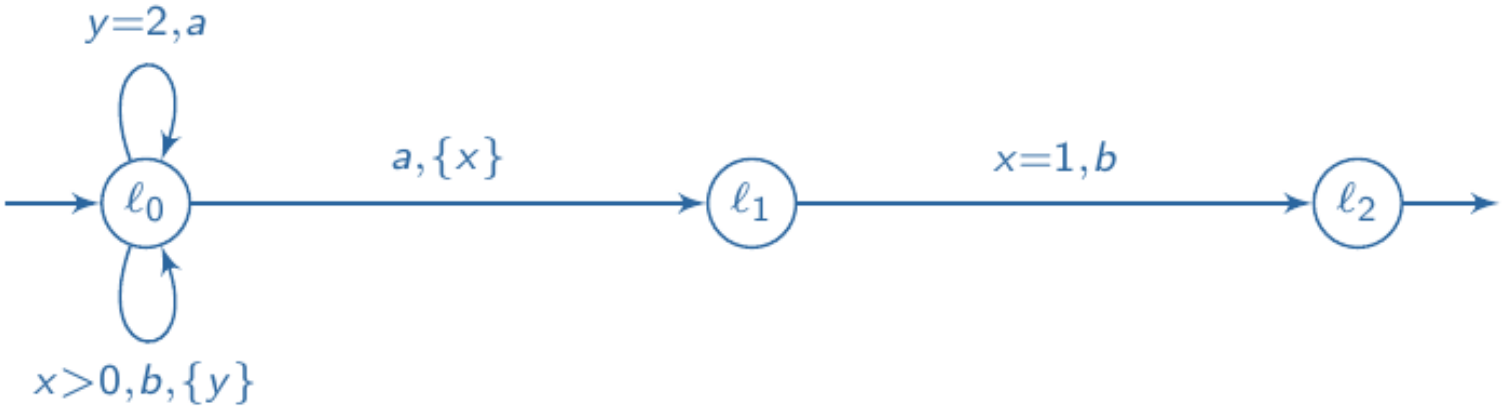
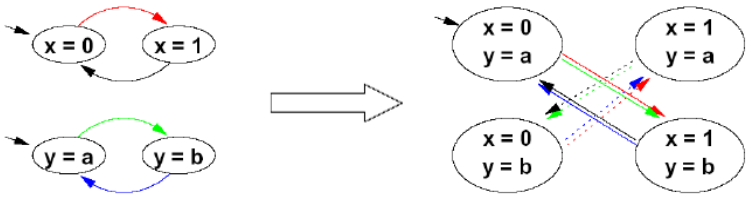


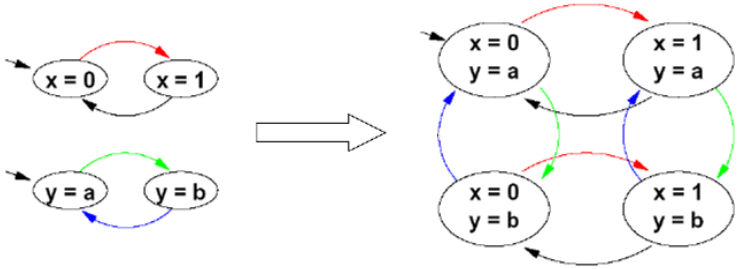

Image by Natalie Bertrand, Inria Rennes Bretagne

# Many variants of TA in the literature (1/2)

- Networks of TA (NTA)
  - Product (parallel composition) of two or more TA with different sets of clocks
  - Different semantics depending on the type of composition
    - Synchronous composition




    - Asynchronous composition




    - Composition with explicit synchronization

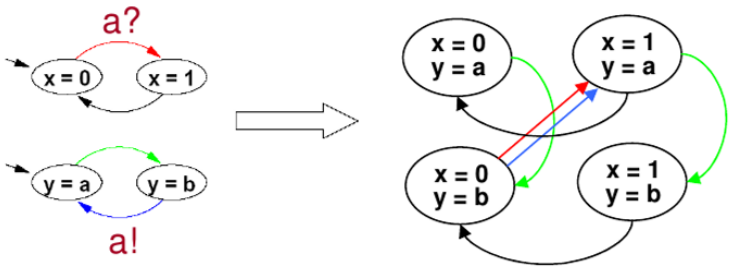



Image by Prof. Marco Di Natale, Scuola Superiore Sant'Anna, Pisa

# Many extensions of TA in the literature (2/2)

- TA with diagonal constraints
  - Guards have syntax $g := x \sim c \mid x - y \sim c \mid g \wedge g$ where $x, y \in X$, $\sim \in \{<, \leq, =, >, \geq\}$, and $c \in \mathbb{N}$ (i.e., a guard is a conjunction of atomic constraints of the form $x \sim c$ and $x - y \sim c$)
- TA with epsilon transitions
  - Epsilon transitions are silent or unobservable transitions
- ...

# Region partitioning (1/2)

- $\forall\, g \in G$, let $[\![g]\!]$ be the set of valuations $\{v \in \mathbb{R}^X_{\geq 0} \mid v \models g\}$
- $\forall\, Y \subseteq X$, let $[Y \leftarrow 0]v$ be the valuation such that
  $[Y \leftarrow 0]v(x) = 0$ if $x \in Y$ and $[Y \leftarrow 0]v(x) = v(x)$ if $x \in X \setminus Y$

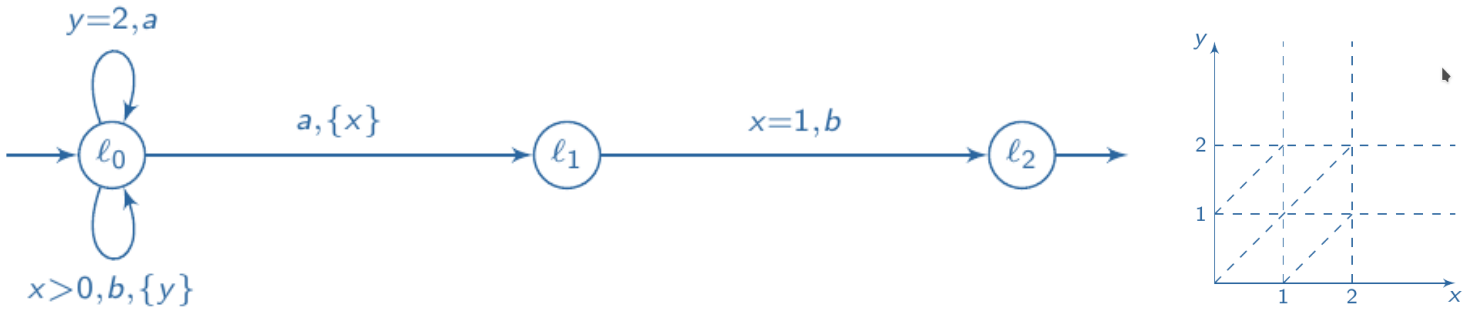
- A finite partition $\mathcal{R}$ of $\mathbb{R}^X_{\geq 0}$ is a **set of regions** (for the set of guards $G$) if:
  1. $\forall\, g \in G$ and $\forall\, R \in \mathcal{R}$, $[\![g]\!] \subseteq R$ or $[\![g]\!] \cap R = \varnothing$
  2. $\forall\, R, R' \in \mathcal{R}$, if $\exists\, v \in R$ and $t \in \mathbb{R}_{\leq 0}$ with $v + t \in R' \Rightarrow$
     $\Rightarrow \exists\, t' \in \mathbb{R}_{\geq 0}$ with $v' + t' \in R'$ $\forall\, v' \in R$
  3. $\forall\, R, R' \in \mathcal{R}$, $\forall\, Y \subseteq X$, if $R_{[Y \leftarrow 0]} \cap R' \neq \varnothing \Rightarrow R_{[Y \leftarrow 0]} \subseteq R$
     (where $R_{[Y \leftarrow 0]}$ is the region obtained from $R$ by resetting clocks in $Y \in X$)

- $\mathcal{R}$ defines an **equivalence relation** over valuations
  - An equivalence class is termed a **region**

- If two valuations are equivalent, then their future behaviors are equivalent
  - condition 1: two equivalent valuations satisfy the same clock constraints
  - condition 2: elapsing of time does not distinguish two equivalent valuations
  - condition 3: resetting clocks does not distinguish two equivalent valuations

# Region partitioning (2/2)

- Two valuations $v$ and $v'$ with $v, v' \in \mathbb{R}_{\geq 0}^{X}$ are equivalent if $\forall\, x, y \in X$:
  - $v(x) > M \Leftrightarrow v'(x) > M$ where $M$ is the **maximal constant** for all clocks
  - $v(x) \leq M \Rightarrow \lfloor v(x) \rfloor = \lfloor v'(x) \rfloor \wedge (\{v(x)\} = 0 \Leftrightarrow \{v'(x)\} = 0)$
    where $\{v(x)\}$ is the fractional part of $\{v(x)\}$
  - $v(x) \leq M \wedge v(y) \leq M \Rightarrow (\{v(x)\} \leq \{v(y)\} \Leftrightarrow \{v'(x)\} \leq \{v'(y)\})$
- The partition is compatible with conditions 1, 2, and 3
- Diagonal-free region partitioning for 2 clocks and maximal constant 2




- Shape of **bounded** regions with 2 clocks and no diagonal constraints:

Images by Natalie Bertrand, Inria Rennes Bretagne

# Region graph

- Is a finite automaton whose states are the regions and whose transitions are:
  - $R \xrightarrow{\tau} R'$ if $\exists\, v \in R,\ v' \in R',\ \tau \in \mathbb{R}_{\geq 0}$ such that $v' = v + \tau$
    (in this case, $R'$ is termed time successor of $R$)
  - $R \xrightarrow{Y} R'$ if $R_{[Y \leftarrow 0]} \subseteq R'$
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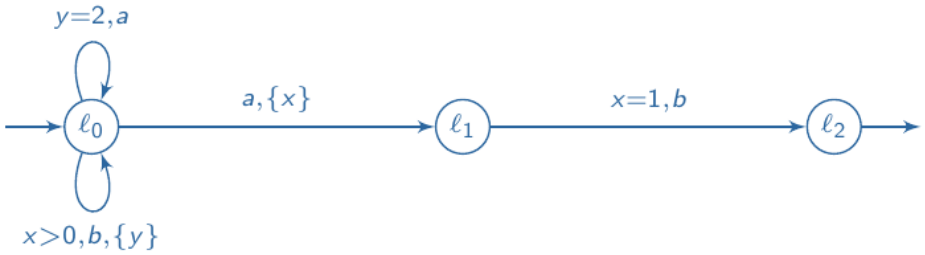


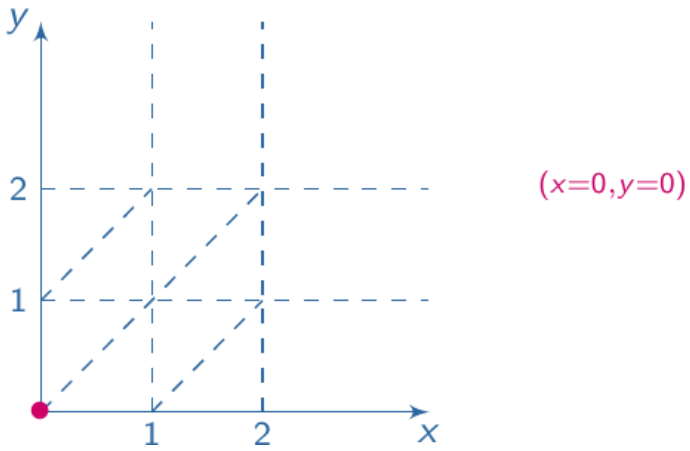


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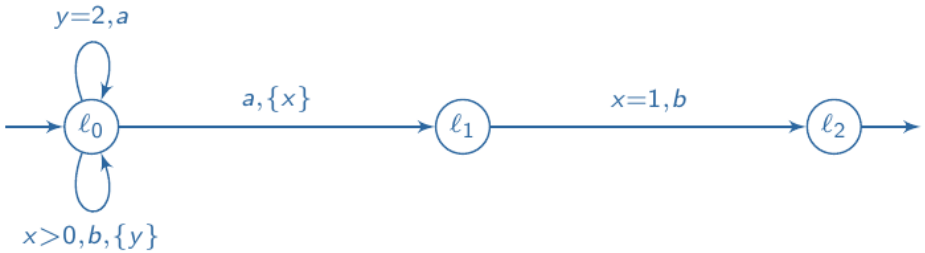


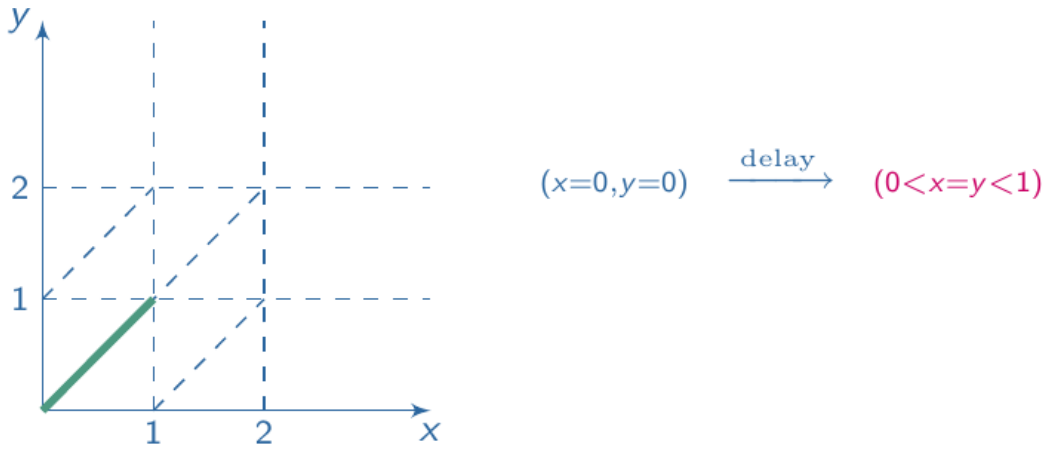


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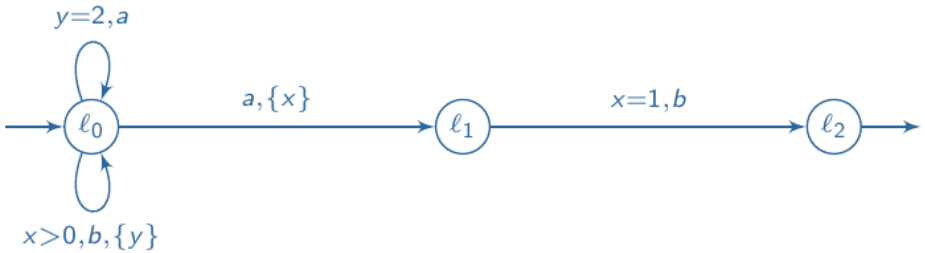


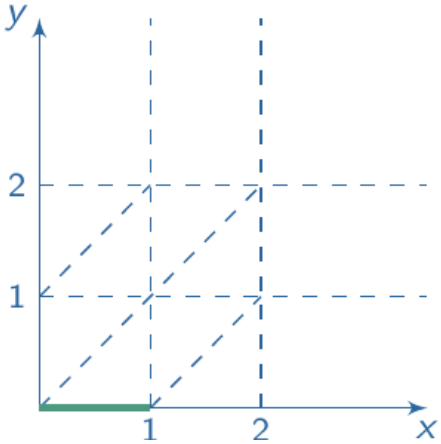


$(x{=}0, y{=}0) \xrightarrow{\text{delay}} (0{<}x{=}y{<}1) \xrightarrow{y:=0} (0{<}x{<}1, y{=}0)$

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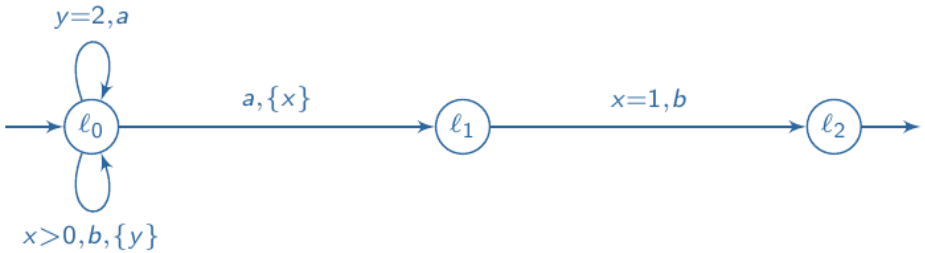


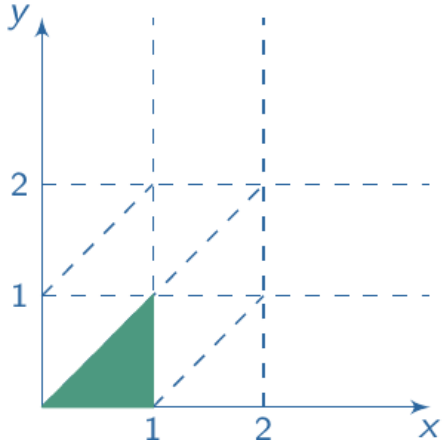


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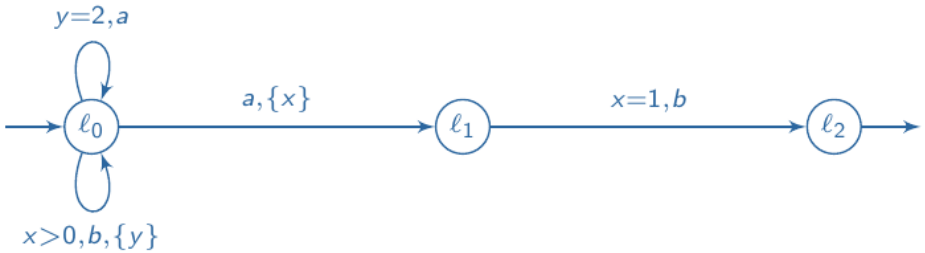


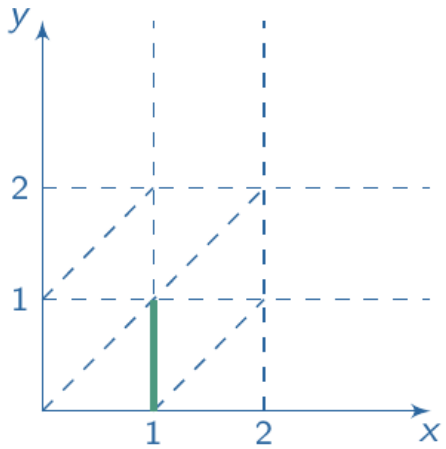


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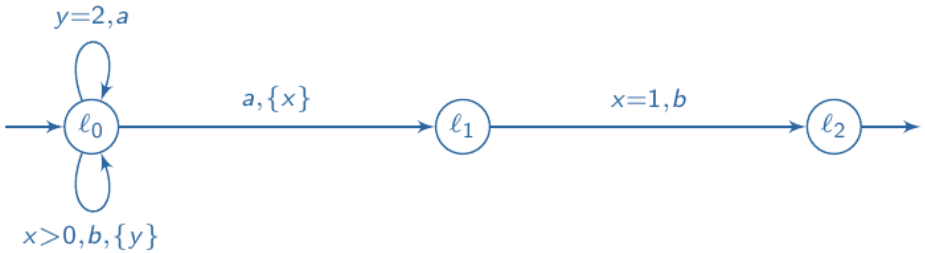


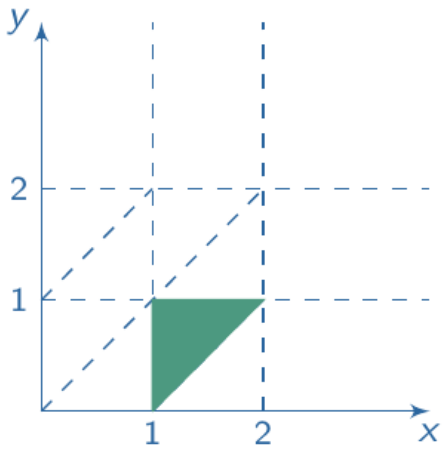


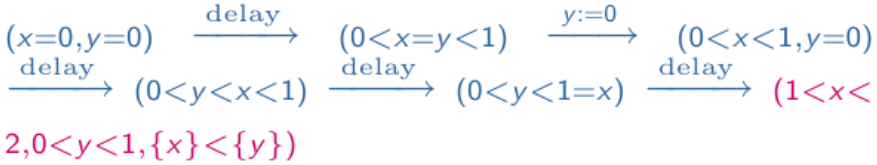


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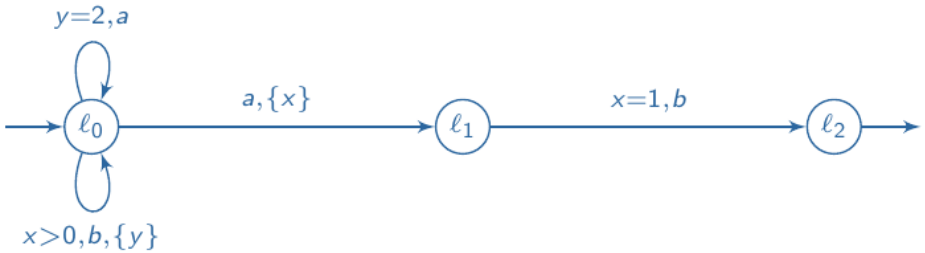


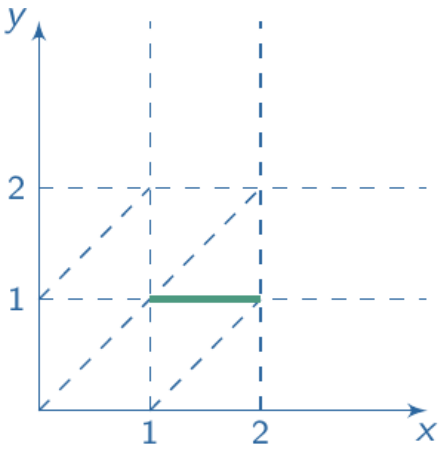


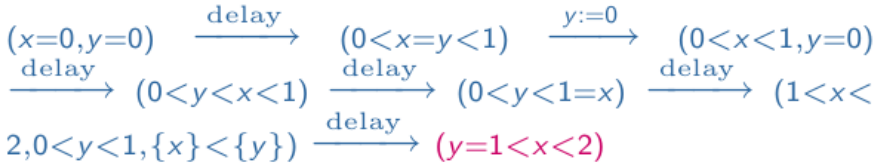


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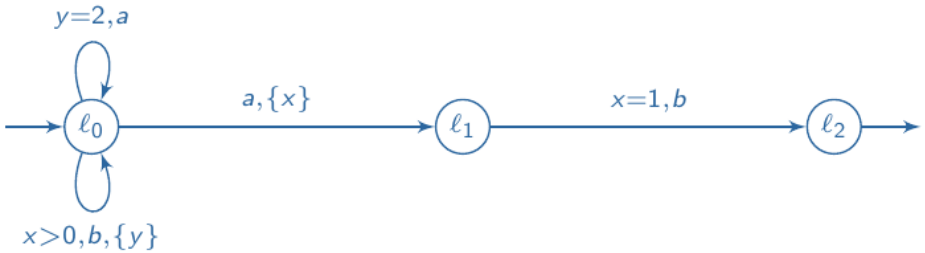
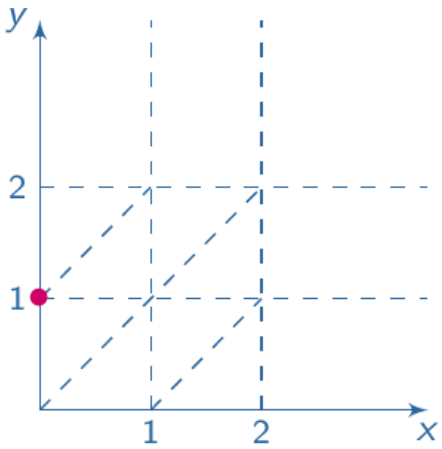


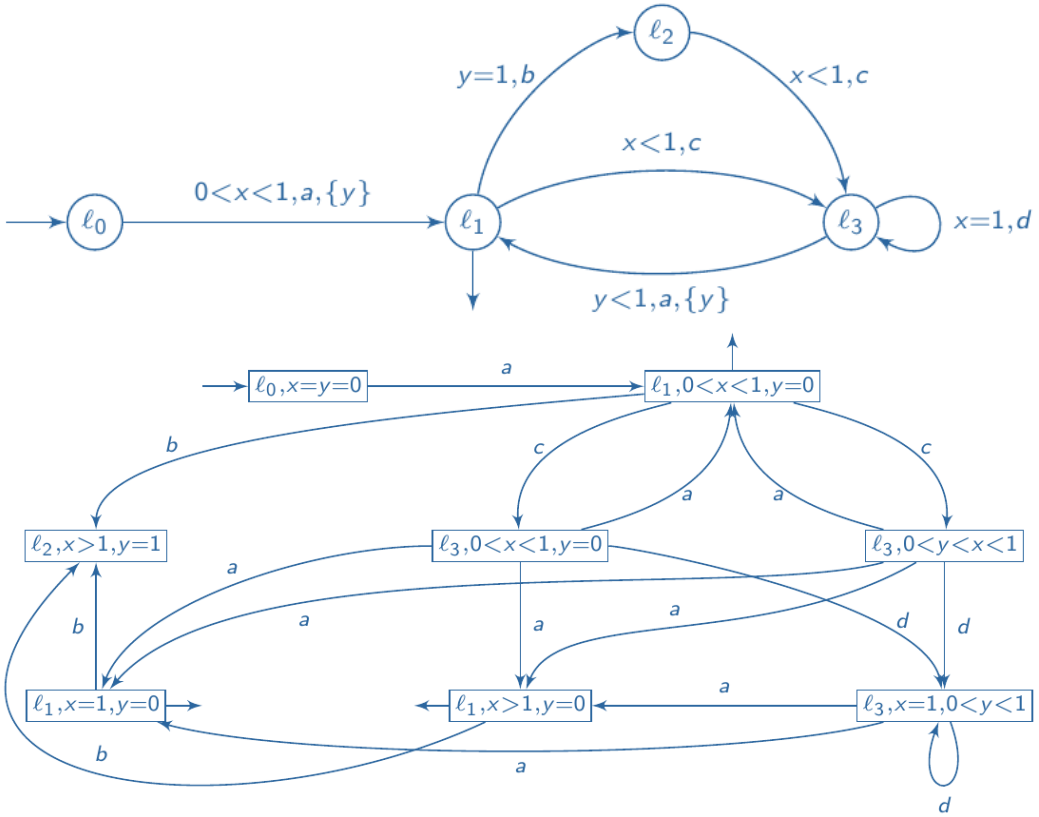




$(x{=}0, y{=}0) \xrightarrow{\text{delay}} (0{<}x{=}y{<}1) \xrightarrow{y:=0} (0{<}x{<}1, y{=}0)$
$\xrightarrow{\text{delay}} (0{<}y{<}x{<}1) \xrightarrow{\text{delay}} (0{<}y{<}1{=}x) \xrightarrow{\text{delay}} (1{<}x{<}$
$2, 0{<}y{<}1, \{x\}{<}\{y\}) \xrightarrow{\text{delay}} (y{=}1{<}x{<}2) \xrightarrow{x:=0} (x{=}0, y{=}1)$

Images by Natalie Bertrand, Inria Rennes Bretagne

# Region automaton

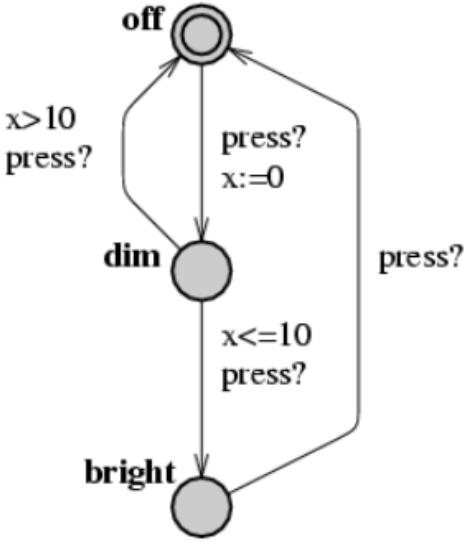
- Is a finite automaton whose set of states is $L \times \mathcal{R}$ and whose transitions are:
  - $(l, R) \xrightarrow{a} (l', R')$ if $\exists\, l \xrightarrow{g,a,Y} l'$ is a transition in the TA with $R \subseteq [\![g]\!]$ and $R \xrightarrow{Y} R'$ is a transition in the region graph
  - $(l, R) \xrightarrow{\tau} (l', R')$ if $R \xrightarrow{\tau} R'$ is a transition of the region graph
- An example




Images by Natalie Bertrand, Inria Rennes Bretagne

# Zone graph

- The number of regions is **exponential** in the number of clocks and the value of the maximal constants in the guards!
- The **zone graph** provides a more efficient encoding
  - Zones are DBMs representing the union of regions
- An example: a TA and its zone graph



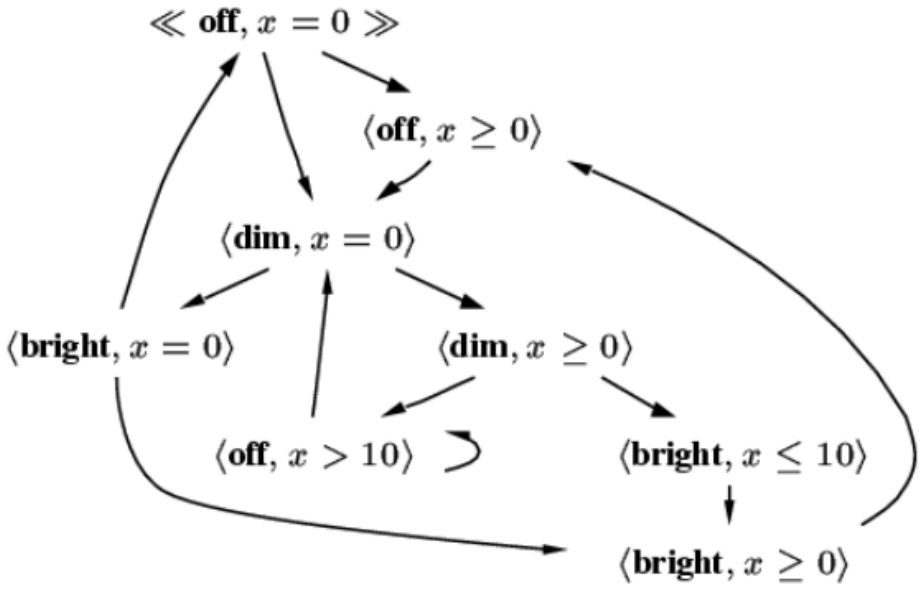




Image by Prof. Marco Di Natale, Scuola Superiore Sant'Anna, Pisa